It is done with the help of three foot screws.

1. The bubble tube is kept parallel to any two foot screws. The foot screws are turned both inwards or outwards until the bubble is central.
2. Now the telescope is rotated to 90° so that it lies over the third foot screw.
3. The third screw is turned until the bubble is central.
4. Now, the telescope is rotated back to its old position. The procedure in steps1 to 3 is repeated till the bubble remains central in all position.
5. The bubble should remain central if the instrument is in proper adjustment.

c. Elimination of parallax

It is the process of making the image of an object to exactly fall on the plane of crosshairs (diaphragm). Unless parallax is eliminated, accurate sighting is impossible. It consists of focusing the eyepiece and objective of level.

i. Focusing the eyepiece

The operation is done to make the crosshairs to appear distinct and clearly visible. The steps involved in this operation are:

1. The telescope is directed towards sky (or) a sheet of white paper is held in front of the objective.
2. The eyepiece is moved in or out till the crosshairs appear distinct.

ii. Focusing the objective

The operation is done to bring the image of the object in the plane of cross hairs. The steps involved in this operation are.

i. The telescope is directed towards the staff

ii. The focusing screw is turned until the image appears clear and sharp.

2. Permanent Adjustment

These are the adjustments which are made to establish fixed relationships between the fundamental parts or fundamental lines of a leveling instrument.

The fundamental lines of a leveling instrument are:

- Axis of bubble tube
- Axis of telescope
- Line of collimation
- Vertical axis

Permanent adjustments once done may keep the instrument in perfect condition for a long time.

Activity 2

Prepare a report about different types of levelling instruments.

3.3.7 Levelling staff

Levelling staff is a graduated rod of rectangular cross section. It is used for the measurement of vertical distance between the line of sight and the point on which the staff is held. It is also known as level rod.

The levelling staff is usually made of best quality seasoned teak wood or aluminium channel. The length of levelling staff in common use are 2 m, 3 m, 4 m and 5 m.

Graduations of the staff give the distances from the bottom of the cap. Each meter is divided into 20 divisions. The thickness of each division is 5 mm.

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